

Practice Problems

1. Design a Student class that represents a student with the following attributes and methods:

Attributes: name, age, grade

Methods:

- a) `__init__(self, name, age, grade)`: Initializes a student object with the given name, age, and grade.
- b) `get_info(self)`: Prints the student's name, age, and grade.
- c) `is_passing(self)`: Returns True if the student's grade is above or equal to 60, False otherwise.

2. Implement a Stack class that represents a stack data structure with the following methods:

- a) `__init__(self)`: Initializes an empty stack.
- b) `push(self, item)`: Pushes an item onto the top of the stack.
- c) `pop(self)`: Removes and returns the item at the top of the stack.
- d) `is_empty(self)`: Returns True if the stack is empty, False otherwise.
- e) `size(self)`: Returns the number of items in the stack.

#Hint: You can use a list as the stack. So your constructor will look somewhat like this-

```
def __init__(self):
```

```
    list_stack=[]
```

3. Design a class so that the following code produces the given output:

#Write your class code here

```
b=Bankaccount(10008753, 25000)
```

```
print(b.account_no)
```

```
print(b.balance)
```

```
b.deposit(5000)
```

```
print(b.balance)
```

```
b.withdraw(10000)
```

```
print(b.balance)
```

Output

10008753

25000

30000

20000

4. Design a class so that the following code produces the given output:

#Write your class code here

```
Rect=Rectangle(10,20)
```

```
print(Rect.height)
```

```
print(Rect.width)
```

```
x=Rect.get_area()
```

```
print(x)
```

Output

10

20

200